GG-C-439b
JULY 18, 1957
SUPERSEDING
Int. Fed. Spec. GG-C-00439a(GSA-FSS)
June 8, 1956 and
Fed. Spec. GG-C-439
May 23, 1946

FEDERAL SPECIFICATION

# CLIP, SUTURE (METAL)

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.**—This specification covers metal suture clips to be used for suturing wounds.

**1.2 Classification.**

**1.2.1** *Type and sizes.*—Clips covered by this specification shall be of the Michel type, and of the following sizes, as specified (see 6.3):

| *Size* | *Length* |
|---|---|
| 1 | 11 millimeters |
| 2 | 14 millimeters |
| 3 | 16 millimeters |
| 4 | 18 millimeters |
| 5 | 22 millimeters |

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

**2.1** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

*Federal Specifications:*

GG-I-526 — Instruments, Dental and Surgical; General Specifications.
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676 — Boxes, Set-Up, Paperboard.

*Federal Standard:*

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

*Military Specifications:*

MIL-P-116—Preservation, Methods of.
MIL-B-10377—Box, Wood, Cleated, Veneer, Paper Overlaid.



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MIL–L–10547 — Liners, Case, Waterproof.

*Military Standards:*

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Material.**—Suture clips shall be made of nickel-silver conforming to class 4, composition S of Federal Specification GG–I–526.

**3.2 Description.**—Suture clips shall be one-piece construction, extra rerolled to gage. The points shall be sharp, evenly tempered, not less than $\frac{1}{16}$-inch long, suitably shaped for maximum strength, and flared at approximately a 45° angle.

**3.3 Dimensions.**—The clips shall conform to the dimensions of table I; tolerance shall be ± 5 percent.

TABLE I.

| Size | Length | Width | Thickness |
|---|---|---|---|
| | *Millimeters* | *Inch* | *Inch* |
| 1 | 11 | 0.106 | 0.012 |
| 2 | 14 | .106 | .013 |
| 3 | 16 | .109 | .014 |
| 4 | 18 | .109 | .015 |
| 5 | 22 | .112 | .016 |

**3.4 Temper.** — Shall be commercial soft temper, and shall provide the proper degree of toughness and rigidity throughout. Processing shall be such that the clips will take the temper required for satisfactory service.

**3.5 Health hazard.**—Suture clips shall not constitute a health hazard to personnel when used as intended. If suture clips are chemically treated, the contractor shall submit a certificate of quality for each lot, with each delivery, showing results of tests performed to confirm that clips will have no deleterious effect on patient's skin. Certificate of quality shall be forwarded by the contractor to the procuring activity (see 6.7).

**3.6 Workmanship.**—Suture clips shall be free from roughness, raggedness, burrs, and other imperfections which will affect their serviceability or necessitate any special treatment or cleaning prior to sterilization for use.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

**4.1 Sampling.**—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.3). In addition, sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

**4.2 Inspection.**—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery as specified (see 6.3). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

**4.3 Tests.**

**4.3.1** *Corrosion-resistance.*—The suture clips taken for test shall, before testing, be scrubbed with soap and water, rinsed thoroughly, and dried. After cleaning, they shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for 24 hours. There shall be no evidence of corrosion as a result of this test.



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4.3.2 *Chemical analysis.*—Upon request, the manufacturer shall furnish millings or drillings for chemical analysis to determine conformity of the material with specification requirements. They shall weigh not less than 4 ounces and shall be clean and free from oil, dirt, grit, or other foreign matter. They shall be taken at slow speed and shall not be oxidized. The manufacturer shall guarantee the sample millings to be of the same material and taken from the stock of which the particular parts were manufactured. In lieu of the sample millings or drillings, a spectrographic analysis, or a certificate of quality, showing results of tests performed to indicate that material is as specified in 3.1 may be furnished.

## 5. PREPARATION FOR DELIVERY

### 5.1 Packaging.

5.1.1 *Unit of issue.*—One hundred clips, of the size specified, constitute one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE II.—*Packaging quantities*

| Unit package | Intermediate package | Shipping container |
|---|---|---|
| 1 unit | 50 units | 1,000 units |

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Twenty-five clips shall be assembled on a wire holder made of brass or nickel silver. Four holders (1 unit) shall be packaged in a kraft paper envelope of suitable weight. Envelope shall be adequately packaged in accordance with method IA–8 of Military Specification MIL–P–116.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level B.*

5.1.4.1 *Unit package.*—Twenty-five clips shall be assembled on a wire holder made of brass or nickel silver. Four holders (1 unit) shall be packaged in a kraft paper envelope of suitable weight. Envelope shall be adequately secured to prevent accidental opening.

5.1.4.2 *Intermediate package.*—Intermediate package shall be as specified in 5.1.3.2.

5.1.5 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

### 5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to



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Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

*Note.*—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.*—The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Unit package.*—Each unit package and bag (when used) shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments.

## 6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 **Intended use.** — Suture clips are intended for use with Michel clip removing forceps.

6.3 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and length required (see 1.2.1).
(c) Sampling required (see 4.1).
(d) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

Level B packaging is intended to provide economical but limited protection, and should be specified only when it is determined the item will be held in covered storage approximately 12 months from date of initial packaging.

6.4 **Storage surveillance.**—Items packaged in accordance with level B requirements must be inspected to determine condition when not used within the time period indicated. Items not used within the time period specified must be repackaged in accordance with level B requirements in this specification or with level A requirements if storage beyond an additional year is anticipated.

6.5 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.



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6.6 This specification covers the following items appearing in the Armed Services Medical Stock List:

6515–375–4100—CLIP, SUTURE, Michel, 11 mm., 100s.
6515–375–4105—CLIP, SUTURE, Michel, 14 mm., 100s.
6515–375–4110—CLIP, SUTURE, Michel, 16 mm., 100s.
6515–375–4115—CLIP, SUTURE, Michel, 18 mm., 100s.

6.7 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification or which are commercially available, but is intended to cover the types, etc., normally procured to meet Federal Government requirements.

6.8 **Transportation description.**—Transportation description applicable to this item is:

Surgical instruments, not otherwise indexed by name.
*Carload minimum weight* 10,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 10,000 pounds.

6.9 **For Military Medical Supply Agency procurement.**—A certificate of quality required if suture clips are chemically treated (see 3.5), shall be submitted to Head, Quality Control Division, Technical Department, Military Medical Supply Agency, 84 Sands Street, Brooklyn 1, New York.

**Notice.** — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

**MILITARY INTERESTS:**
**Army—M**
**Navy—Md**
**Air Force.**

Copies of this specification may be purchased for 10 cents each.



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